

Munaza Ashraf

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Education

University of Engineering & Technology (UET), Taxila, Pakistan

2020 - 2024

Bachelor of Science (Honours) in Software Engineering - CGPA: 3.94/4.0

Relevant Courses: Machine Learning, Artificial Neural Networks, Calculus and Analytical Geometry, Linear Algebra and Differential Equations, Probability and Statistics, Digital Image Processing, Computer Vision, Data Structures & Algorithms, Operating Systems, Database Management Systems, Cloud Computing

Research Interests

[1] Machine Learning [2] Multimodal and Generative AI [3] Foundation and World Models [4] Computer Vision and 3D Vision [5] Embodied AI and Robot Learning [6] Efficient Model Training [7] Distillation and Inference

Research Direction: Developing efficient world and multimodal models for embodied AI, with a focus on transferring spatial, temporal, and physical reasoning into compact models through knowledge distillation and efficient training and inference under strict latency, memory, and compute constraints.

Technical Skills

Programming: Python, C++, CUDA, Go, SQL

Machine Learning: PyTorch, Transformers, Diffusion Models, LLMs, VLMs, Computer Vision, Multimodal AI, LoRA/PEFT

Training and Inference: Distributed Training and Inference, FSDP, Tensor Parallelism, Context Parallelism, Quantization, Mixed Precision, FlashAttention, KV Caching, PyTorch Compilation

Model Domains: Image, Video, 3D, Speech, Generative and Multimodal Models

Systems: vLLM, FastAPI, Docker, Linux, AWS, RunPod

Profiling and Evaluation: CUDA Profiling, Latency and Throughput Benchmarking, VRAM and GPU Utilization, Model Quality Evaluation, Experiment Tracking

Experience

Machine Learning Engineer | Applied ML Research | Runware | London, England

Nov 2024 - Present

- Integrate, optimize, and serve image, video, audio, 3D, LLM/VLM, and multimodal models in production, with inference improvements of up to **2× across selected workloads**.
- Built and optimized inference systems serving **1,000+ models with GPU memory management**, model loading and unloading, batching, concurrency control, and workload isolation.
- Research and implement model acceleration techniques involving **attention optimization, KV and QK caching, diffusion caching, context parallelism, distributed execution, compilation, quantization and GPU aware scheduling**.
- Benchmark models across H100, H200, B300 and RTX PRO 6000 hardware to study compute, memory and communication tradeoffs.
- Worked with high-volume enterprise workloads including **Higgsfield, Freepik, Wix, and Mayflower**, focusing on throughput, concurrency, GPU utilization, and production reliability.

Product Engineer | Faz Australia | Australia [product based]

Sep 2024 - July 2025

- **Founding engineer for MedTalk**, contributing across machine learning, backend systems and production ML infrastructure.
- **Fine tuned** and deployed speech, **medical NLP and language models** for real time transcription and structured clinical generation.

- Reduced post VAD speech processing latency from approximately **150 ms P50 to approximately 50 ms**.
- Built real time WebSocket based inference systems and **deployed ML workloads across AWS and GPU infrastructure**.
- Developed **RAG and multimodal AI workflows** for domain specific medical intelligence.

Computer Vision Engineer | VisionRD | Pakistan

June 2024 - Nov 2024

- Developed **computer vision and generative AI systems** for industrial environments.
- Built **automated worker activity and assembly validation systems** for Thal Engineering.
- Developed **defect detection systems** for KIA Motors to **automate visual quality inspection**.
- Built and optimized real-time computer vision pipelines for industrial monitoring, assembly validation, and visual defect detection, focusing on robust and low-latency production inference.

Academic Project

PsycheSync: AI Powered Mental Health and Wellness Platform | Final Year Project

2023 - 2024

- Developed an **AI driven mHealth platform** for mental health support, including mood tracking, journaling, and psychological assessments.
- Fine tuned and integrated **LLaMA 2 and OpenAI language models** on domain specific data for empathetic and context aware responses.
- Designed a **semantically controlled community environment** with automated moderation for safer user interactions.
- Integrated the AI pipeline into a **Flutter based application** for real time interaction.

Professional Projects

Multimodal Model Inference & Optimization Platform | Runware | runware.ai

Nov 2024 - July 2025

- Built a **unified execution layer** for heterogeneous image, video, audio, 3D, LLM/VLM, and multimodal architectures with **routing, batching, streaming, hot-swapping, and VRAM-aware scheduling**.
- Integrated **caching, sparse and parallel attention, compilation, and automated profiling** to compare latency, throughput, VRAM, and output-quality trade-offs across models and hardware.

Multi-Architecture LoRA Training Platform | Runware | runware.ai/lora-training

May 2026 - July 2026

- Engineered modular **LoRA/QLoRA training** for **diffusion, LLM, and VLM** architectures with dataset preprocessing, captioning, checkpointing, validation, and memory-efficient training.
- Implemented adapter evaluation, dynamic LoRA loading, merging, weighted composition, and production hot-swapping.

Padelix | Real-Time Multi-Camera Sports Perception

March 2026 - Aug 2026

- Designed a **video analytics** pipeline for **player and ball tracking**, clothing-based re-identification, court calibration, homography, and rally/highlight extraction from fixed multi-camera footage.
- Focused on **robust perception** under occlusion, identity persistence, camera timing mismatch, and fast post-match inference.

MedTalk | Real-Time Multimodal Clinical AI | medtalk.co

Sep 2024 - July 2025

- Built and deployed **low-latency speech** and **language-model pipelines** for live clinical transcription and structured generation, including WebSocket streaming and model serving.
- Optimized the speech pipeline from about 150 ms P50 post-VAD latency to about 50 ms while integrating **fine-tuned speech, medical NLP, and LLM** components.

Unilever x PSL 2025 | Real Time Broadcast Logo Detection and Sponsorship Analytics

April 2025 - June 2025

- Built a real time computer vision pipeline using **YOLOv8 and OpenCV** to detect, track, and classify sponsor logos by visibility quality while measuring screen time across live broadcast footage.
- Aggregated detections with match context such as overs, camera angles, and events to generate automated exposure reports and **ROI based sponsorship analytics**.

Automated Lumber Takeoff and Cost Estimation from Construction Drawings

July 2025 - Feb 2026

- Built a multi stage computer vision pipeline using **YOLO for object detection, SAM for segmentation, and Detectron2 based models for region level analysis**, extracting lumber types, dimensions, quantities, and structural context from construction drawings..
- Integrated **Gemma and Qwen vision language models** for semantic interpretation of drawing annotations and material context, then deployed the pipeline as a **FastAPI cloud service** for automated bill of materials and cost estimation.

Real Time Optical Fiber Intrusion Detection using CNN LSTM

July 2024 - Sep 2024

- Built a real time signal classification pipeline using **FFT, wavelet transforms, filtering, and noise reduction** to extract temporal and frequency domain features from optical fiber pulse data.
- Trained and deployed a **CNN LSTM model** for intrusion detection within a continuous Python streaming pipeline supporting low latency inference and automated alerts.

Relevant Internships

NESCOM | Machine Learning Intern | Islamabad, Pakistan

March 2023 - Oct 2023

- Developed an **Ad Detector & Classifier System** using **deep-learning models** to detect and categorize online advertisements, with **data collection, preprocessing, feature engineering, and model evaluation**.
- Worked on machine-learning tasks involving signal processing and pattern recognition, with emphasis on model accuracy, reliability, and efficient preprocessing.

Bytewise Limited | Deep Learning Intern | Islamabad, Pakistan

May 2022 - Sep 2022

- Designed and **trained deep-learning models** for computer vision and NLP using **TensorFlow and PyTorch**; built custom datasets and data-augmentation pipelines.
- Implemented **classification and detection workflows** and evaluated model performance on real-world tasks.

Selected Honors & Technical Community

- **Silver Medal**, UET Taxila; **ranked 2nd in overall session throughout 4 years**.
- **Google Developer Student Club**, UET Taxila | Active Member, 2020 – 2024.
- **Microsoft Learn Student Ambassadors** | Active Member, 2020 – 2024.
- **IEEE, UET Taxila Chapter** | Event Host, 2020 – 2024.
- **Umeed-e-Subh UET Taxila** | General Secretary | 2020 - 2024

References

Engr. Wajahat Abbas

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Kash Pande

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